

SOLUTION:**1. b) 18**

$$1 + 2 + 3 + 4 = 10$$

$$\text{and } 1 + 3 + 5 + 8 = 17$$

$$\text{Similarly, } 1 + 4 + 6 + 9 = 20$$

2. c) 108

$$(4 + 8) \times 9 = 108$$

$$(5 + 4) \times 12 = 108$$

3. c) 10

For first triangle,

$$10 - 4 = 6$$

$$18 - 10 = 8$$

$$18 - 4 = 14$$

For second triangle,

$$14 - 8 = 6$$

$$22 - 14 = 8$$

$$22 - 8 = 14$$

For third triangle,

$$11 - 5 = 6$$

$$15 - 11 = 4$$

$$15 - 5 = 10$$

4. d) 22

$$(2 \times 2 - 1) = 3$$

$$\text{and } (5 \times 4 - 5) = 15$$

$$(5 \times 5 - 3) = 22.$$

5. c) 32

$$(5 \times 6 \times 4) / 10 = 12$$

$$\text{and } (6 \times 7 \times 5) / 10 = 21$$

$$\text{Therefore } (4 \times 8 \times 10) / 10 = 32$$

6. c) 108

$$13 + 15 = 28$$

$$36 + 54 = 90$$

$$\text{Therefore, } 45 + 63 = 108$$

7. d) 11

$$(3 \times 4 - 8) = 4$$

$$(2 \times 5 - 4) = 6$$

$$(4 \times 5 - 9) = 11$$

8. c) 25

$$\sqrt{4 \times 9} = 6$$

and

$$\sqrt{9 \times 16} = 12$$

Therefore

$$\sqrt{16 \times ?} = 20$$

$$? = 25$$

9. c) 37

$$(3 \times 3) + (5 \times 6) = 39$$

$$\text{and } (4 \times 4) + (5 \times 7) = 51$$

$$\text{Therefore, } (3 \times 4) + (5 \times 5) = 37$$

10. d) 9

$$93 - (27 + 3) = 63$$

$$79 - (38 + 4) = 37$$

$$\text{Therefore, } 67 - (16 + X) = 42$$

$$X = 9$$

11. b) 2

Putting the position of the letters in reverse order

$$P = 11, S = 8, V = 5 \text{ and } Y = 2$$

12. a) 9

$$5 \times 8 \times 10 = 400$$

$$\text{and } 6 \times 9 \times 7 = 378$$

$$\text{Therefore } 5 \times 7 \times ? = 315$$

$$? = 9$$

13. c) 4

$$(16)^2 = 256$$

Therefore, $(2)^2 = 4$

14. d) 4/5

From I row, $1 + (1/2) = 3/2$
From II row, $2 + (2/3) = 8/3$
From III row, $3 + ? = 19/5$

$$? = (19/5) - 3$$

$$? = (4/5)$$

15. b) 26

$(2)^2 + (4)^2 = 20$
and $(3)^2 + (9)^2 = 90$
Therefore $(1)^2 + (5)^2 = 26$

16. b) 24

$(7 \times 3) = 21$ and $(9 \times 3) = 27$
and $(4 \times 9) = 36$ and $(2 \times 9) = 18$
Therefore $(9 \times 6) = 54$ and $(4 \times 6) = 24$

17. b) 30

$(16 + 13) = (14 + 15)$
and $(28 + 12) = (10 + 30)$
Therefore $(29 + 16) = (15 + 30)$

18. d) 18

$(3 \times 4) + 3 = 15$
and $(7 \times 5) + 3 = 38$
Therefore $(3 \times 5) + 3 = 18$

19. a) 11

$(15 - 12) + (10 - 9) = 4$
 $(28 - 12) + (16 - 20) = 12$
Similarly, $(23 - 11) + (15 - 16) = 11$

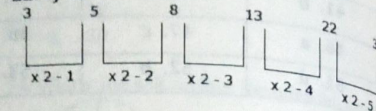
20. d) 15

Sum of numbers in lower half of the circle = 11
 $+ 9 + 3 + 7 = 30$
Sum of numbers in upper half of the circle = ?
 $+ 4 + 5 + 6 = ? + 15$
Upper half = Lower half
 $30 = ? + 15$
 $? = 30 - 15$

$$? = 15$$

21. b) 622

$915 - 364 = 551$
and $789 - 543 = 246$
Similarly, $863 - 241 = 622$

22. c) 39**23. a) 184**

$(12)^2 - (8)^2 = 80$
and $(16)^2 - (7)^2 = 207$
Therefore $(25)^2 - (21)^2 = 184$

24. d) 73

$(2)^3 + (1)^3 + (3)^3 = 36$
and $(0)^3 + (4)^3 + (3)^3 = 91$
Therefore, $(4)^3 + (2)^3 + (1)^3 = 73$

25. d) K₈

$$A_2 \xrightarrow{+2} C_4 \xrightarrow{+2} E_6$$

$$M_5 \xrightarrow{+2} O_9 \xrightarrow{+2} Q_{14}$$

$$G_3 \xrightarrow{+2} I_5 \xrightarrow{+2} K_8$$

How the number is obtained?

$$2 + 4 = 6$$

$$5 + 9 = 14$$

$$\text{Similarly,}$$

$$3 + 5 = 8$$

Therefore, the answer is K₈

26. d) 6

$(5 + 4 + 7)/2 = 8$
 $(6 + 9 + 5)/2 = 10$
 $(3 + 7 + 2)/2 = 6$

27. c) 9

$(12 + 18 + 30)/10 = 6$
 $(16 + 24 + 40)/10 = 8$
Similarly, $(45 + 18 + 27)/10 = 9$

28. c) 41

$(5 \times 3) + 4 = 19$
and $(6 \times 4) + 5 = 29$
Therefore, $(7 \times 5) + 6 = 41$

29. a) 45

$(15 \times 2 - 3) = 27$,
 $(31 \times 2 - 6) = 56$
and $(45 \times 2 - 9) = 81$

30. a) 1

$(2)^2 = 4$
 $(8)^2 = 64$
 $(5)^2 = 25$
 $(1)^2 = 1$

31. b) 7

$(10 + 9 + 14 + 7)/2 = 20$
and $(11 + 16 + 10 + 13)/2 = 25$
Therefore, $(15 + ? + 12 + 10)/2 = 22$

$$\text{Hence } 37 + ? = 44$$

$$? = 44 - 37$$

$$? = 7$$

32. d) 37

$(5 \times 6) + 2 = 32$
 $(7 \times 6) + 2 = 44$
 $(7 \times 5) + 2 = 37$

33. a) 6

$(5 + 3)/2 = 4$
and $(6 + 8)/2 = 7$
Therefore $(8 + 4)/2 = 6$

34. d) 690

$(1)^2 + (5)^2 + (4)^2 + (3)^2 = 51 \times 10 = 510$
and $(3)^2 + (4)^2 + (6)^2 + (2)^2 = 65 \times 10 = 650$
Similarly $(0)^2 + (1)^2 + (2)^2 + (8)^2 = 69 \times 10 = 690$

35. b) 622

$(915 - 364) = 551$
 $(789 - 543) = 246$
 $(863 - 241) = 622$

36. c) 11

$(4 \times 8) + 1 = 33$
 $(5 \times 9) + 1 = 46$
Similarly, $(7 \times 11) + 1 = 78$

37. b) 26

$(2)^2 + (4)^2 = 20$
 $(3)^2 + (9)^2 = 90$
Therefore, $(1)^2 + (5)^2 = 26$

38. b) 6

$3 \times 5 \times 4 = 60$
and $5 \times 7 \times 4 = 140$
Therefore, $4 \times 4 \times ? = 96$

$$? = (96/16) = 6$$

39. c) 6

$(4 \times 7) \% 4 = 7$
and $(6 \times 2) \% 3 = 4$
Therefore, $(6 \times 2) \% 2 = 6$

40. c) 125

All numbers are cubed,
 $(7)^3 = 343$
 $(1)^3 = 1$
 $(3)^3 = 27$
Similarly, $(5)^3 = 125$

41. b) 6

$(4 + 9 + 2) = (3 + 5 + 7) = (8 + 1 + 6)$
Total in each case = 15

42. b) 13

$(15 + 12)/9 = 3$
and $(44 + 28)/9 = 8$
Therefore, $(64 + 53)/9 = 13$

43. b) 38

$9 + (2)^2 = 13$
 $13 + (3)^2 = 22$
 $22 + (4)^2 = 38$

44. c) 86

$(30 - 24) \times 8 = 48$
and $(23 - 12) \times 8 = 88$
Therefore, $(92 - 86) \times 8 = 48$

45. c) 11

$(6 + 5) - (7 + 4) = 0$
and $(7 + 6) - (8 + 4) = 1$
Therefore $(11 + 2) - (2 + 0) = 11$

46. a) 64

The numbers are squared in ascending order

In first circle,

$$(3)^2 = 9$$

$$(4)^2 = 16$$

$$(5)^2 = 25$$

$$(6)^2 = 36$$

In second circle,

$$(4)^2 = 16$$

$$(5)^2 = 25$$

$$(6)^2 = 36$$

$$(7)^2 = 49$$

In third circle,

$$(6)^2 = 36$$

$$(7)^2 = 49$$

$$(8)^2 = 64$$

$$(9)^2 = 81$$

47. c) 16

$$(4 \times 6) \% 3 = 8$$

$$(6 \times 10) \% 5 = 12$$

$$(4 \times 8) \% 2 = 16$$

48. b) 5

$$(7 \times 3) + 8 = 29$$

$$(4 \times 3) + 7 = 19$$

$$(5 \times ?) + 6 = 31$$

$$? = 5$$

49. a) 94

$$(4)^2 + (2)^2 + (1)^2 = 21$$

$$\text{and } (5)^2 + (3)^2 + (8)^2 = 98$$

$$\text{Therefore } (6)^2 + (7)^2 + (3)^2 = 94$$

50. b) 130

$$(5 \times 6 \times 8) + (7 \times 4 \times 9) = 492$$

$$\text{and } (7 \times 5 \times 4) + (6 \times 8 \times 9) = 572$$

$$\text{Therefore } (4 \times 3 \times 5) + (7 \times 2 \times 5) = 130$$

51. d) 99

$$21 + 31 = 52$$

$$\text{and } 39 + 46 = 85$$

$$\text{Therefore, } 16 + 83 = 99$$

52. b) 3

$$(18 \times 12)/3 = 72$$

$$\text{and } (32 \times 16)/4 = 128$$

$$\text{Therefore, } (24 \times 14)/? = 112$$

$$(336/?) = 112$$

$$? = (336/112)$$

$$? = 3$$

53. c) 38

$$(0 + 2 + 6 + 4) - 2 = 10$$

$$\text{and } (6 + 2 + 10 + 8) - 2 = 24$$

$$\text{Therefore, } (4 + 14 + 12 + 10) - 2 = 38$$

54. b) 64

$$(1)^3 = 1, (2)^3 = 8, (3)^3 = 27$$

$$\text{Therefore, } (4)^3 = 64$$

55. c) 49

$$(5)^2 = 25$$

$$(6)^2 = 36$$

$$(4)^2 = 16$$

$$(7)^2 = 49$$

Hence the number 49 will replace the question mark